

Modelling Financial Markets

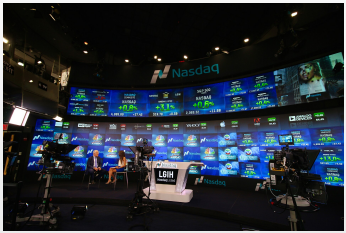
Chapter 10: Market Microstructure



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What We Will Explain: from Orders to Prices and Costs



Nasdaq MarketSite, Times Square, New York: a studio, not a trading floor

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- **Questions**: when computers match orders in microseconds
  - what does it cost to trade?
  - how do prices absorb information?
- The rules of the matching engine (priority, tick size, auctions) shape prices and costs Gould et al. (2013)
- Models: order book, spread (Roll, Corwin–Schultz, EDGE), Amihud ratio, Glosten–Milgrom, Kyle, Almgren–Chriss
- Applications: stress episodes, crypto, the Bucharest Stock Exchange (BVB), the speed race
- Worked exercises: Seminar 10
- Textbooks: Hasbrouck (2007), Bouchaud, Bonart, Donier & Gould (2018), Franke, Härdle & Hafner (2019)

Reading the Market Series and Units

- BET means Bucharest Exchange Trading, the main BVB stock index; TR means Total Return, with reinvested dividends
- For intraday charts, r is the change in the log close from one bar to the next

The Evidence: SPY, BVB, Crypto and the VIX

Series	Content	Period
SPY, 5-minute bars	ETF (exchange-traded fund) on the S&P 500; regular session 09:30–16:00 New York time; 1 477 full days	2020–2026
Bitcoin, 5-minute bars	24 hours a day, 7 days a week; 208 596 bars	2024–2026
US stocks, daily	SPY, Apple, Microsoft, Nvidia, JPMorgan, GameStop, Strategy	2016–2026
BVB stocks, daily	Ten blue chips and the ETF on the BET (the BVB benchmark index); values in lei converted to USD at the reference rate of the National Bank of Romania (BNR)	2016–2026
Crypto-assets, daily	Bitcoin, Ethereum, Solana, XRP, Dogecoin, Cardano, Litecoin	2016–2026
VIX, daily	Index of the 30-day implied volatility of the S&P 500, computed from option prices	2020–2026

- Daily data: open, high, low, close and volume; prices adjusted for dividends and splits; days without trades removed
- Order books in the section *The Limit Order Book* and the bid-ask bounce in the section *The Bid-Ask Spread* are **simulated**, and labelled as such

Notation: One Symbol, Several Models

Symbol	Meaning, by model
T	Roll in finite samples: number of price changes (77 per SPY day); Almgren–Chriss, Avellaneda–Stoikov: the horizon
α	Corwin–Schultz: spread part of the log range; PIN: probability of an information event; VECM: adjustment coefficients; simulated book: limit-order rate
β	Corwin–Schultz: sum of the squared one-day ranges; Kyle: the insider's trading intensity; VECM: cointegrating vector
γ	Roll: autocovariances γ_k; Corwin–Schultz: squared two-day range; Almgren–Chriss: permanent impact; Avellaneda–Stoikov: risk aversion
λ	Kyle: price move per share of net order flow; Almgren–Chriss: risk aversion; PIN: Poisson rate in $\pi(n;\lambda)$; Hawkes: intensity $\lambda(t)$
μ	Glosten–Milgrom: share of informed traders; PIN: arrival rate of informed orders; simulated book: market-order rate; Hawkes: baseline rate
η	Abdi–Ranaldo and EDGE: log mid-range η_t; Almgren–Chriss: temporary impact
δ, ε	δ: PIN bad-news probability, cancellation rate, quote distance (Avellaneda–Stoikov); ε: Roll news shock, PIN uninformed rates $\varepsilon_b, \varepsilon_s$, order sign (propagator)

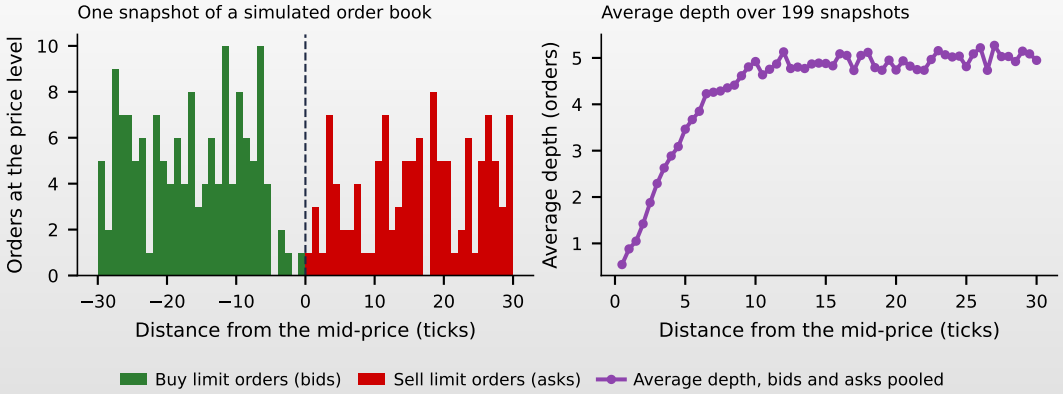
Liquidity and the Limit Order Book: Definitions

- **Liquidity**: trading a quantity quickly at a price close to fair value; no single number captures it
  - **Tightness** (the spread), **depth** (size needed to move the price), **resiliency** (speed of recovery) Kyle (1985); **immediacy** Harris (2003)
- **Limit order book** (LOB): all standing limit orders, by price and time of arrival Gould et al. (2013)
- At time t: best bid $b_t <$ best ask a_t; mid-price $m_t = (a_t + b_t)/2$; relative spread $(a_t - b_t)/m_t$ in basis points (1 bp $= 0.01\%$)
  - **Depth**: quantity waiting at a price level; **tick size**: smallest price step (USD 0.01 for most US stocks above USD 1)

Orders, Priority and Matching

- **Market order**: immediate execution against the available liquidity, uncertain price; **limit order**: price bounded by the limit, uncertain execution (other order types: ▸ Appendix)
- Resting limit orders supply liquidity; market orders and marketable limit orders consume it
- **Price-time priority**: the better price first, then the earlier order at the same price (a queue); queue position is valuable (case study: the high-frequency trading arms race)
- **Call auctions** at the open and close: orders are pooled and matched at the single price that maximises the executed quantity
- A market buy of size Q walks up the ask levels: it pays the half-spread plus the cost of the deeper levels

A Simulated Order Book (1/2)



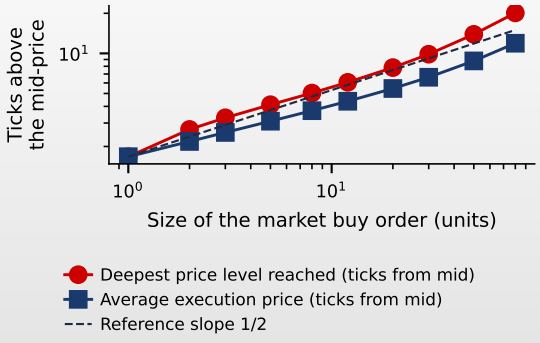
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A Simulated Order Book (2/2)

- **Simulated**: zero-intelligence order flow Farmer, Patelli & Zovko (2005), limit orders, market orders and cancellations arriving at random (rates: Appendix)
  - Left: one snapshot (bids green, asks red); right: mean depth by distance from the mid
- Mean spread 3.23 ticks, median 3, above 6 ticks 10% of the time; depth 0.8 orders at 1 tick, 4.8 at 10, 5.0 at 25 (199 snapshots)
- Market orders eat the best levels first, so liquidity near the price is scarce Tóth et al. (2011); real books show the same rise away from the mid Bouchaud, Bonart, Donier & Gould (2018)

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The Price Impact of a Market Order



- Market buys of 1 to 80 units sent against each simulated snapshot; log-log axes
- Marginal price: a 1-unit order reaches 1.7 ticks above the mid, an 80-unit order 20.1 ticks
- **Concave** impact: slope 0.53, close to 1/2
- Average execution price: slope 0.44; 80 units pay 11.9 ticks per unit on average
- Doubling the order: cost per unit $\times 2^{\mathbf{0.44}} \approx 1.36$, not 2; total cost (size $\times$ cost per unit) $\times 2^{\mathbf{1+0.44}} \approx 2.7$
- The impact is concave because book depth grows with distance from the mid
- Large orders split over time (metaorders) show a similar law (section *Price Impact and Optimal Execution*)

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Why Is There a Spread?

- A liquidity provider buys at the bid and sells at the ask
- The spread pays three costs Huang & Stoll (1997)
  - **Order processing**: fees, technology, clearing
  - **Inventory**: holding an unwanted position exposes the provider to price risk
  - **Adverse selection**: some counterparties know more Bagehot (1971)
- The provider incurs losses when trading with informed traders and must recover them from uninformed traders (section *Information and Prices*)

Quoted, Effective and Realised Spread (1/2)

- Three ways to measure what a round trip costs, all relative to the mid-price
  - p_t: price of the trade at time t; m_t: the mid-price just before the trade
  - q_t: trade direction, $+1$ if the buyer initiated the trade (it hit the ask), -1 if the seller did
- **Quoted** spread: the width of the posted quotes, from quotes only
  - $S_t^Q = (a_t - b_t)/m_t \geq 0$
- **Effective** spread: what the trade actually paid, twice its signed distance from the mid
  - $S_t^E = 2q_t(p_t - m_t)/m_t$
  - Multiplying by q_t makes the cost positive on both sides: a buy above the mid and a sell below it both give $S_t^E > 0$
  - Round-trip measure: the one-way cost is $S_t^E/2$; $S_t^E < S_t^Q$ when trades execute inside the quotes

Quoted, Effective and Realised Spread (2/2)

- **Realised** spread: what the liquidity provider keeps once the price has moved
  - $S_t^R = 2q_t(p_t - m_{t+\Delta})/m_t$
  - $m_{t+\Delta}$: the mid-price a time Δ after the trade (e.g. 5 minutes); revenue before fees and inventory costs
  - $S_t^R < 0$: the provider lost, because the price kept moving in the direction of the trade
- **Price impact**: the part of the effective spread that the provider loses to the later price move
  - $S_t^E - S_t^R = 2q_t(m_{t+\Delta} - m_t)/m_t$
  - The lasting move of the mid after the trade, usually attributed to adverse selection, though inventory and order-flow pressure also contribute
- Quoted spread: quotes only; effective and realised spread: trades, the prevailing quotes and the trade direction, signed for example by the rule of Lee & Ready (1991)
  - Lee–Ready: a trade above the mid is a buy, below it a sell; at the mid, the tick test: above the last different trade price is a buy, below it a sell
- Without such data we **estimate** the spread from prices only: the next slides

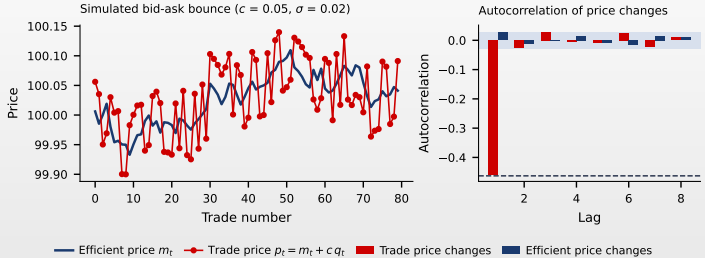
The Roll Model (1/2)

- Idea Roll (1984): the spread can be read from the way trade prices bounce between bid and ask
  - $t = 1, \ldots, n$: successive trades; prices in USD (or log prices)
- **Efficient price** m_t: a random walk, it moves only on news
  - $m_t = m_{t-1} + \varepsilon_t$
  - ε_t: news shock, i.i.d. (independent, identically distributed) with mean 0 and variance σ^2
- **Trade price** p_t: the efficient price plus or minus half the spread
  - $p_t = m_t + c\, q_t$
  - $c > 0$: half-spread, the unknown parameter; the spread is $s = 2c$
  - $q_t = +1$ (buy at the ask) or -1 (sell at the bid), each with probability 1/2, i.i.d. over time and independent of ε

The Roll Model (2/2)

- Price change between two trades: news plus the bounce
  - $\Delta p_t = p_t - p_{t-1} = \varepsilon_t + c(q_t - q_{t-1})$
- Its variance and its covariance with the previous change
  - $\mathrm{Var}(\Delta p_t) = \sigma^2 + 2c^2, \qquad \mathrm{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$
  - The common term q_{t-1} enters Δp_{t-1} with $+c$ and Δp_t with $-c$: a jump up to the ask tends to be followed by a fall to the bid
- **Roll estimator**: solve $\mathrm{Cov} = -c^2$ for c, with the covariance replaced by its sample value
  - $\hat{s} = 2\hat{c} = 2\sqrt{-\widehat{\mathrm{Cov}}(\Delta p_t, \Delta p_{t-1})}$
  - $\widehat{\mathrm{Cov}}$: sample first-order autocovariance of the $n-1$ price changes; defined only if it is negative
- The bid-ask bounce creates negative autocorrelation even when the efficient price is a random walk

The Bid-Ask Bounce (Simulated)



- $n = 5\,000$ simulated trades with $c = 0.05$, $\sigma = 0.02$; light-blue band $\pm 1.96/\sqrt{n}$: the 95% band for the autocorrelations of white noise; dashed line: theoretical first-order autocorrelation
- True spread $2c = 0.10$, Roll estimate $\hat{s} = 0.0998$; lag-1 autocorrelation -0.458 against the theoretical $\mathrm{Cov}/\mathrm{Var} = -c^2/(\sigma^2 + 2c^2) = -0.463$
- No autocorrelation beyond lag 1: the market is efficient, yet trade prices are not a random walk

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The Roll Estimator in Finite Samples (1/2)

- On real data the sample covariance is often **positive**, so $\hat{s}$ does not exist: SPY 5-minute bars, positive on 41% of days
  - **Sampling error**: the model is true, but T is small and c^2 is small relative to σ^2
  - **Misspecification**: trends after news, orders split into many trades, stale prices, bars instead of trades
- Notation for the next slide Harris (1990)
  - $\gamma_k = \mathrm{Cov}(\Delta p_t, \Delta p_{t-k})$: autocovariance at lag k; in the Roll model $\gamma_0 = \sigma^2 + 2c^2$ (the variance) and $\gamma_1 = -c^2$
  - T: the number of price changes (78 closes in a day give $T = 77$)
  - $\hat{\gamma}_1$: its sample value from the T price changes, each centred on their sample mean (demeaned)
  - $\mathbb{E}$: expected value over repeated samples; Φ: distribution function of the standard Normal distribution

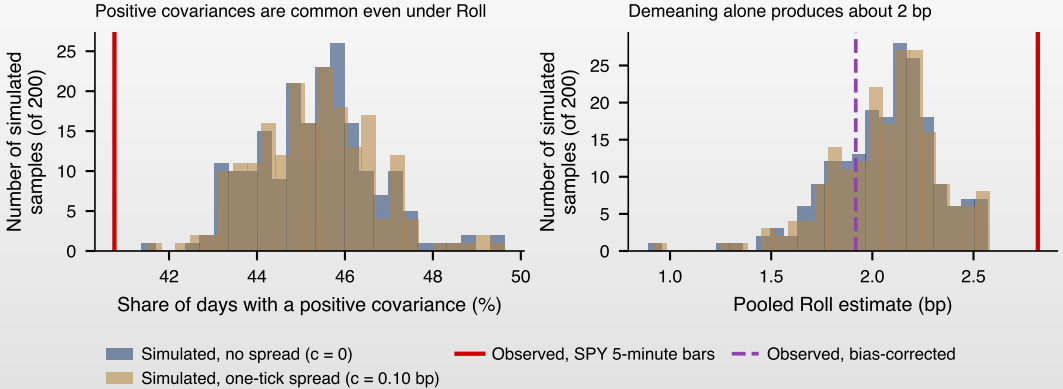
The Roll Estimator in Finite Samples (2/2)

- **Bias**: subtracting the sample mean pushes $\hat{\gamma}_1$ down
  - $\mathbb{E}[\hat{\gamma}_1] \approx \gamma_1 - (\gamma_0 + 2\gamma_1)/T$
  - Negative even when there is no spread ($c = 0$): Roll then finds a spread that is not there
- **Noise**: the sampling variance of $\hat{\gamma}_1$ shrinks only as $1/T$
  - $T\,\mathrm{Var}(\hat{\gamma}_1) \to \gamma_0^2 + 3\gamma_1^2 \quad (T \to \infty)$
- **Probability of a wrong sign**: $\hat{\gamma}_1$ is approximately Normal, so
  - $P(\hat{\gamma}_1 > 0) \approx \Phi(\sqrt{T}\,\gamma_1/\gamma_0)$
  - Close to 1/2 (a coin toss) whenever $c^2 \ll \sigma^2/\sqrt{T}$ ($\ll$: much smaller than): the spread is small relative to the news noise
- Delta-method standard error and the daily-data case: Appendix; seminar A1–A2 derive them step by step

Demeaning Pulls the Wrong-Sign Share Below One Half

- Bartlett approximation of the sampling standard deviation: $s_\gamma = \sqrt{(\gamma_0^2 + 3\gamma_1^2)/T}$; the form $\Phi(\sqrt{T}\,\gamma_1/\gamma_0)$ of the previous slide neglects $3\gamma_1^2$
- The Normal approximations so far centre $\hat{\gamma}_1$ at γ_1; demeaning moves the centre to the biased mean $\gamma_1 - (\gamma_0 + 2\gamma_1)/T$
  - Same Normal approximation, shifted centre, Bartlett standard deviation:
    $$P(\hat{\gamma}_1 > 0) \approx \Phi\Big(\frac{\gamma_1 - (\gamma_0 + 2\gamma_1)/T}{\sqrt{(\gamma_0^2 + 3\gamma_1^2)/T}}\Big)$$
- No spread ($c = 0$, so $\gamma_1 = 0$): the argument is $(-\gamma_0/T)/(\gamma_0/\sqrt{T}) = -1/\sqrt{T}$
  - $P \approx \Phi(-1/\sqrt{T}) < 1/2$: positive covariances on fewer than half of the days, from demeaning alone
- SPY, $T = 77$, one-tick spread: $\gamma_0 \approx 83.9$ bp^2 (the 9.2 bp bar standard deviation, squared), $\gamma_1 = -c^2 \approx -0.011$ bp^2, shift $(\gamma_0 + 2\gamma_1)/T \approx 1.09$ bp^2
  - $\Phi(-0.115) \approx 45.4\%$, inside the simulated band of the next slides; the unshifted formula gives 49.96%, outside it

SPY: Sampling Error or Misspecification? (1/2)



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SPY: Sampling Error or Misspecification? (2/2)

- Each day has 77 five-minute returns; a one-tick spread means a half-spread $c = 0.10$ bp, against a bar standard deviation of 9.2 bp
- The band comes from 200 simulated samples of the same size under the Roll model
- With a one-tick spread, $\hat{\gamma}_1 > 0$ on 45.3% of days (band: [43.0, 48.1]), close to $\Phi(-1/\sqrt{77}) \approx 45.5\%$ (demeaning alone)
- In the data the share is 40.8%, below the band: the negative autocorrelation is stronger than a one-tick bid-ask bounce

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SPY: Demeaning Bias and the Corrected Roll Spread

- No spread ($c = 0$): the Roll estimator from the autocovariance averaged over all days is still 2.06 bp [1.50, 2.53]
  - In autocovariance units (bp^2): the mean over days $\bar{g}_1 = -1.99$ behind the observed 2.82 bp; about 54% of it (-1.07) is demeaning bias
  - Solving the bias formula for γ_1, with $\hat{\gamma}_0 \approx 83.9$: $(\bar{g}_1 + \hat{\gamma}_0/T)/(1 - 2/T) \approx -0.92$, so the bias-corrected spread is $2\sqrt{0.92} \approx 1.92$ bp: transient price pressure, not the bounce
  - Spreads do not subtract ($2\sqrt{\cdot}$ is concave), squares do: $2.06^2 + 1.92^2 \approx 2.82^2$

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Spreads from Daily Ranges: Corwin–Schultz (1/2)

- Idea: the daily high is usually a buy at the ask, the low a sell at the bid; the high-low range = volatility + spread Corwin & Schultz (2012)
- The volatility part grows with the length of the interval, the spread part does not: compare one day with two days
- Notation
  - H_t, L_t: high and low price of day t; $H_{t,t+\mathbf{1}}$, $L_{t,t+\mathbf{1}}$: high and low over the two days t and $t+1$
  - $\ln(H/L)$: the log range, a proxy for volatility; $j = 0, 1$ runs over the two days
- Two building blocks
  - $\beta = \sum_{j=\mathbf{0}}^{\mathbf{1}} \left[\ln \frac{H_{t+j}}{L_{t+j}}\right]^{\mathbf{2}}, \qquad \gamma = \left[\ln \frac{H_{t,t+\mathbf{1}}}{L_{t,t+\mathbf{1}}}\right]^{\mathbf{2}}$
  - β: sum of the two squared one-day ranges; γ: squared two-day range
  - Without a spread, $\gamma \approx \beta$ on average (variance adds up over two days); the spread inflates β twice, γ only once

Spreads from Daily Ranges: Corwin–Schultz (2/2)

- Solving the two moment equations for the spread gives first α, then the spread estimate $\hat{S}$
  - $\alpha = \dfrac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\dfrac{\gamma}{3 - 2\sqrt{2}}}, \qquad \hat{S} = \dfrac{2(e^{\alpha} - 1)}{1 + e^{\alpha}}$
  - α: the part of the log range due to the spread, after the volatility part is removed; the constant is $3 - 2\sqrt{2} = (\sqrt{2} - 1)^2$, so $\gamma = \beta$ gives $\alpha = 0$ (next slide)
  - $\hat{S}$: relative spread, a fraction of the price ($\times 10^4$ for bp); $\hat{S} \approx \alpha$ for small α
- Reading a value
  - $\hat{S} = 0.001$: a round trip costs about 10 bp; $\alpha < 0$ (volatility larger than the model expects) gives $\hat{S} < 0$
  - Negative two-day estimates are set to 0, then averaged over all pairs of days
- For daily data the paper adjusts for overnight moves (Appendix)

Two Limit Cases of the Corwin–Schultz Formula

- Write the constant as $(\sqrt{2}-1)^2$; with $\sqrt{2\beta}-\sqrt{\beta}=(\sqrt{2}-1)\sqrt{\beta}$ the formula becomes
  - $\alpha=\dfrac{\sqrt{\beta}-\sqrt{\gamma}}{\sqrt{2}-1}$ (the same α, rewritten)
- **Test 1, pure volatility**: one-day log ranges 0.01 and 0.01, two-day range $0.01\sqrt{2}$ (variance adds up)
  - $\beta=2\times 0.01^2=\gamma$, so $\alpha=0$ and $\hat{S}=0$: no spread is found where there is none
- **Test 2, pure spread**: the price only bounces between a bid and an ask 1% apart, so every range, one-day or two-day, is 0.01
  - $\beta=2\times 0.01^2$, $\gamma=0.01^2$: $\alpha=(\sqrt{2}-1)\,0.01/(\sqrt{2}-1)=0.01$, and $\hat{S}\approx 0.01$, i.e. 100 bp: the whole range is spread

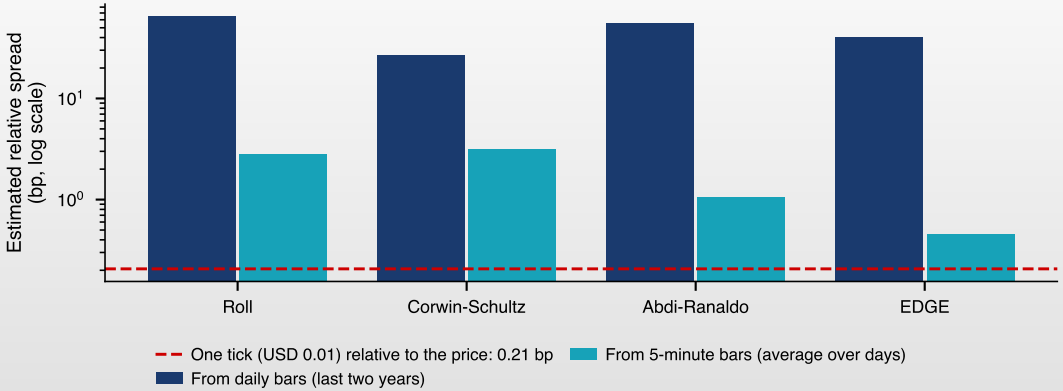
Abdi–Ranaldo and EDGE (1/2)

- **Abdi–Ranaldo** Abdi & Ranaldo (2017): the close sits at the bid or the ask, the middle of the day's range proxies the mid-price
  - c_t: log closing price of day t; $\eta_t = (\ln H_t + \ln L_t)/2$: log mid-range of day t
  - $c_t - \eta_t$: distance of the close from today's mid-range; $c_t - \eta_{t+1}$: from tomorrow's
- The product of the two distances is, on average, a quarter of the squared spread
  - $\hat{S}^2 = 4\,\mathbb{E}\,[(c_t - \eta_t)(c_t - \eta_{t+1})]$
  - $\mathbb{E}$ is replaced by the sample mean over days; news moves η_{t+1} independently of the bounce at c_t, so the volatility terms average out
  - S: relative spread, a fraction of the price; a negative mean product gives $\hat{S} = 0$

Abdi–Ranaldo and EDGE (2/2)

- **EDGE** (Efficient Discrete Generalized Estimator) Ardia, Guidotti & Kroencke (2024): uses open, high, low and close (OHLC) prices
  - Two moment conditions, each a product of log-returns between the open, the close and η_t (exact form: Appendix)
  - They give two unbiased estimates $\hat{e}_1$, $\hat{e}_2$ of S^2, with sampling variances v_1, v_2
- Minimum-variance combination (diagonal weighting): each estimate is weighted by the **other**'s variance
  - $\hat{S}^2 = \dfrac{v_2 \hat{e}_1 + v_1 \hat{e}_2}{v_1 + v_2}$
  - The noisier estimate (larger v) gets the smaller weight; equal weights 1/2 when $v_1 = v_2$
- Common to both: only OHLC bars are needed; a negative mean moment gives $\hat{S} = 0$; $\sqrt{\hat{S}^2}$ is not exactly unbiased for S
- For crypto-assets, high-low estimators track the time variation of high-frequency liquidity; the Amihud ratio is better for levels Brauneis, Mestel, Riordan & Theissen (2021)

Spread Estimators at Two Frequencies: SPY



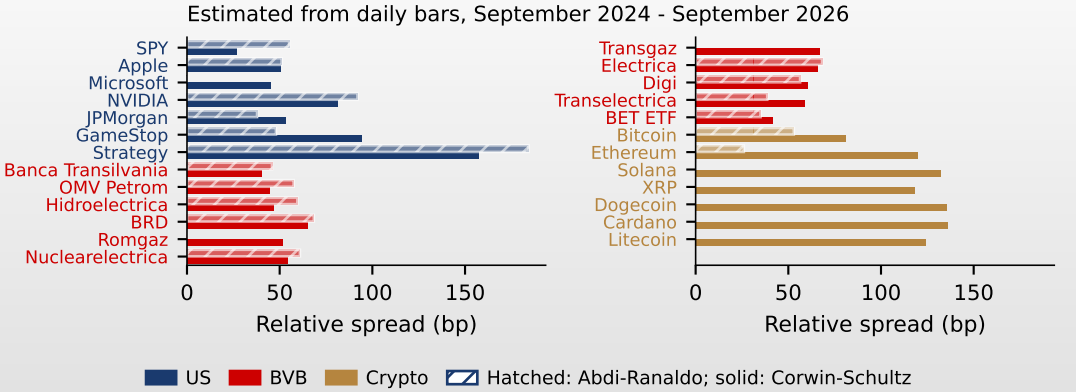
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SPY Spread Estimates: Daily versus Five-Minute Bars

- Daily bars Sep 2024 – Sep 2026; 5-minute bars, 1 477 days (2020–2026)
- One tick (USD 0.01) is 0.21 bp: a floor for the **quoted** spread, not for the effective one
- Daily, in bp: two orders of magnitude above a tick Jahan-Parvar & Zikes (2023)
  - Roll 64.3, Corwin–Schultz 26.5, Abdi–Ranaldo 55.1, EDGE 40.1: volatility dominates
- 5-minute, in bp: EDGE is the closest to a tick
  - Roll 2.82, Corwin–Schultz 3.11, Abdi–Ranaldo 1.06; EDGE 0.46 [0.42, 0.49]
  - EDGE per day from its 5-minute bars, negative days set to 0, then averaged; interval: 95% percentile bootstrap over days (1 000 resamples)
- A tick is a plausibility check, not a validation

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Spread Estimates Across 25 Assets



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Why Daily Spread Proxies Rank BVB Next to US Large Caps

- Daily bars, Sep 2024 – Sep 2026; BVB stocks in lei, prices adjusted for dividends and splits
- Median Corwin–Schultz: US 53, BVB 54, crypto 124 bp
- BVB blue chips look as liquid as US large caps: implausible, given how little value they trade (next section)
  - Median daily traded value, same window and groups: USD 10 505 million in the US, USD 0.34 million on the BVB, a ratio of about 31 000
- The ranking follows volatility; Abdi–Ranaldo is 0 for 8 of the 25 assets (negative sample mean)
- Levels need an independent benchmark: the effective spread measured from trades Jahan-Parvar & Zikes (2023)

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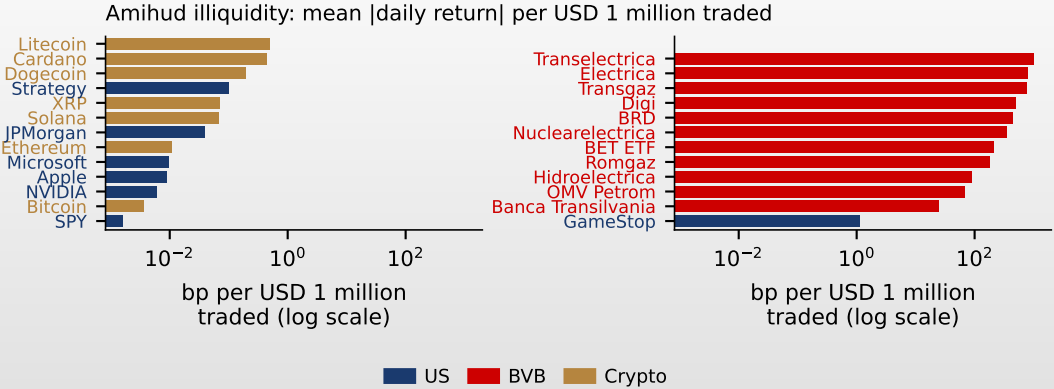
The Amihud Illiquidity Ratio

- Daily price impact per unit of money traded Amihud (2002)
  - $\mathrm{ILLIQ}_i = \dfrac{1}{D_i} \displaystyle\sum_{d=1}^{D_i} \dfrac{|r_{i,d}|}{\mathrm{DVOL}_{i,d}}$
  - i: the asset; $d = 1, \ldots, D_i$: its trading days with nonzero volume; $r_{i,d}$: daily return; $\mathrm{DVOL}_{i,d}$: traded value (price $\times$ volume)
  - $\mathrm{ILLIQ}_i \geq 0$; larger means less liquid: the same traded value moves the price more
- Units here: basis points of price move per USD 1 million traded
- Needs only daily returns and volumes: available for every market and every period
- A low-frequency proxy of the Kyle price-impact coefficient (section *Information and Prices*) Hasbrouck (2009)
- Traded value of BVB stocks: value in lei divided by the BNR reference rate of the day; crypto: traded value in USD summed across exchanges

Reading the Amihud Ratio: Units and Kyle's Lambda

- A toy day: the price moves $|r| = 50$ bp while USD 25 million is traded
  - $|r|/\mathrm{DVOL} = 50/25 = 2$ bp per USD 1 million; ILLIQ_i averages such daily ratios
  - The same day is 200 bp per USD 100 million: divide by 100 to go back to USD 1 million
- The intraday SPY measure later in this section uses bp per USD 100 million and averages 78 bar ratios a day
  - Its mean 1.83 is 0.0183 bp per USD 1 million, still not comparable with the daily SPY ratio (0.0016): a bar carries about 1/78 of the day's value, but its $|r|$ is not 78 times smaller
- Link to Kyle's λ (section *Information and Prices*): the price move per share of net order flow y
  - If all volume were net flow, $\Delta p = \lambda y$ at price p gives $|r|/\mathrm{DVOL} = (\lambda|y|/p)/(p|y|) = \lambda/p^2$
  - Offsetting buys and sells add volume without moving the price, so the ratio is at most λ/p^2 (seminar A6)

Amihud Illiquidity Across 25 Assets



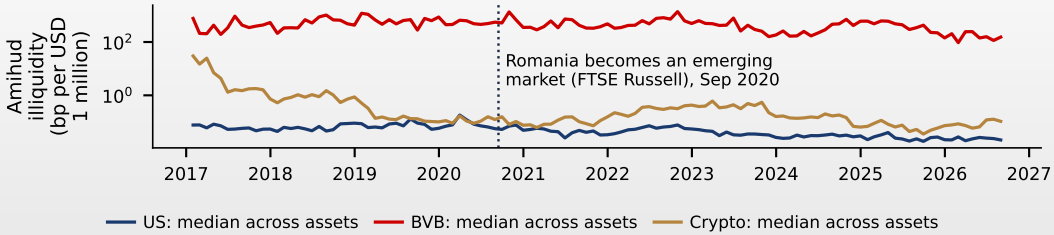
MFM_ch10_amihud

Amihud Ratios: Three Clusters Orders of Magnitude Apart

- Sep 2024 – Sep 2026, log scale; mean over about 500 trading days (730 for crypto)
- SPY 0.0016 bp per USD 1 million (median value USD 40 124 million a day); Bitcoin 0.0037 bp
- Banca Transilvania, the most liquid BVB stock: 24.7 bp
  - about 15 500 times SPY; Transelectrica 1 024.0; the BET ETF 212.9
- Three clusters, separated by orders of magnitude
  - US large caps and top crypto; smaller crypto and GameStop; the BVB
- Daily spread proxies do not separate markets into these three clusters
- A proxy, not a cost you could actually trade at

MFM_ch10_amihud

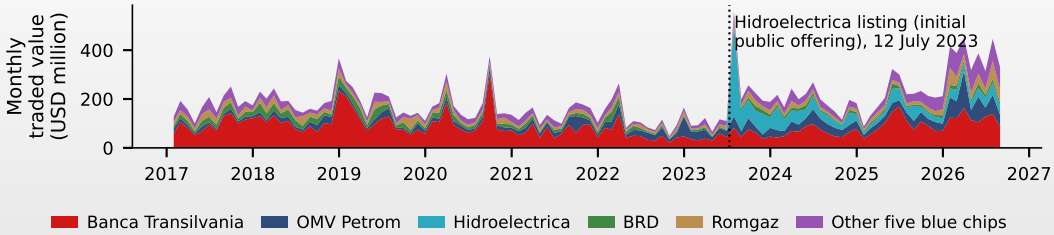
Illiquidity over Time: Three Markets



- Monthly Amihud ratio, median across each group (without SPY and the BET ETF), log scale
- Crypto: from 7.6 bp per USD 1 million (2017) to 0.089 (2026); US: from 0.062 to 0.023
- BVB: from 435 to 169, still about 7 000 times the US median; no break at the move to emerging-market status by the index provider FTSE Russell (September 2020) BVB (2026)

MFM_ch10_amihud

Traded Value of BVB Blue Chips



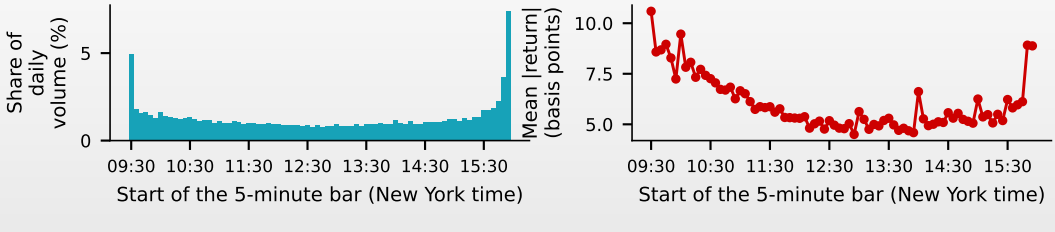
- Chart: monthly traded value; annual totals (sums of the monthly values) of the ten blue chips, USD at the BNR reference rate: 2 098 million in 2017, 1 494 million in 2022, 2 520 million in 2025
- Concentration: Banca Transilvania 44% of the 2025 value, Hidroelectrica (listed July 2023) 13%
- One median SPY day (USD 40 124 million) is about 16 times the 2025 total of the ten blue chips: a position of a few million USD takes days to build or unwind
- The BVB in brief: ▸ Appendix

MFM_ch10_bvb

Is Liquidity Priced?

- **Level**: investors demand a higher return for assets that are costly to trade Amihud & Mendelson (1986)
  - Expected return increases with the spread, concavely: long-horizon investors hold the illiquid assets
- **Time series**: expected market illiquidity raises future returns; an unexpected rise lowers current prices Amihud (2002)
- **Liquidity risk**: stocks that fall when market liquidity dries up earn a premium Pástor & Stambaugh (2003), Acharya & Pedersen (2005)
- Evidence here: monthly market returns fall when illiquidity rises unexpectedly on the BVB, in the US and in crypto (seminar C2)
  - **Unexpected** illiquidity: the residual of an AR(1) fitted to the market's monthly illiquidity, as in Amihud (2002); the monthly return is regressed on last month's illiquidity and this residual (Nov 2016 – Aug 2026)
  - *t*-statistics of the illiquidity-shock coefficient: -4.0 (BVB), -3.8 (US), -4.1 (crypto)

SPY Within the Trading Day



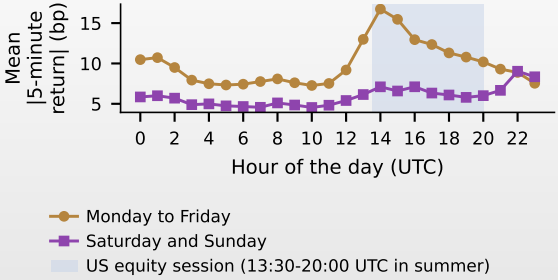
- 78 five-minute bars a day, 1 477 full days (Oct 2020 – Sep 2026); the first bar starts at the opening price
- Volume: first bar 5.0% of the day, a midday bar 0.9%, last 30 minutes 18.6% [18.3, 19.0] against 7.7% if even
- Mean absolute 5-minute return $|r|$: 10.6 bp in the first bar, 5.1 bp at midday, ratio 2.06 [1.98, 2.15] (95% confidence interval, CI, resampling whole days, seminar B1); the spikes at 10:00 and 14:00 coincide with scheduled macroeconomic data releases

MFM_ch10_intraday

Why a U-Shape?

- Documented for stocks of the New York Stock Exchange (NYSE) by Wood, McInish & Ord (1985): returns and volume are highest at the open and the close
- **Open**: overnight news is absorbed; uncertainty is highest
  - Informed traders act early, before the information spreads
- **Clustering**: uninformed traders who can choose the time trade together, and liquidity attracts liquidity Admati & Pfleiderer (1988)
- **Close**: index funds and ETFs trade at the closing price, which is fixed by the closing auction
- Practical rule: an execution algorithm follows the volume curve to reduce impact (section *Price Impact and Optimal Execution*)

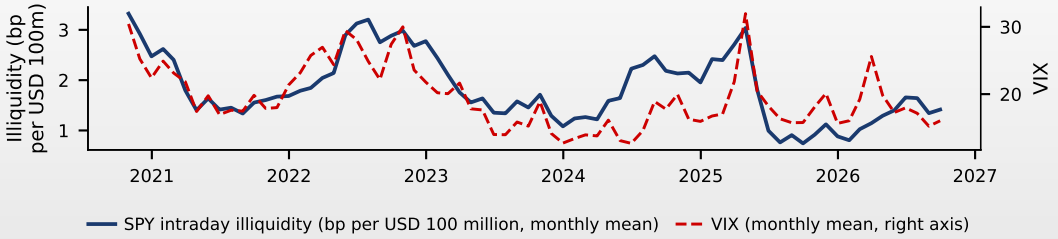
Bitcoin Around the Clock



- Mean absolute 5-minute return by hour of Coordinated Universal Time (UTC), Sep 2024 – Sep 2026, 208 596 bars
- Weekdays: peak 16.7 bp at 14:00 UTC
  - US open, macro releases; low 7.3 bp at 10:00 UTC
- Weekend 5.9 bp against 9.9 bp on weekdays
  - 1.69 times calmer (confidence intervals: seminar B7)
- A 24/7 market still follows the clock of banks, ETFs and traditional investors

MFM_ch10_intraday

Liquidity and Volatility: SPY and the VIX



- Daily intraday illiquidity: mean over the 78 bars of $|r|$ / traded value, bp per USD 100 million (divide by 100 for USD 1 million); monthly means shown
- Correlation with the VIX: 0.58 daily, 0.69 for monthly means; elasticity of log illiquidity on log VIX 0.95 (HAC standard error, robust to heteroskedasticity and autocorrelation: 0.12; fitted relation in seminar B4)
- April 2025: 3.04 against a sample mean of 1.83: liquidity dries up when investors most need it, so liquidity risk is correlated with market risk Acharya & Pedersen (2005)

MFM_ch10_intraday

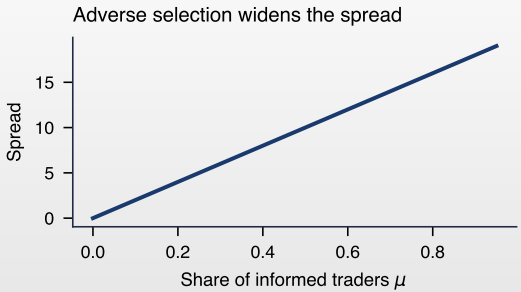
The Glosten–Milgrom Model

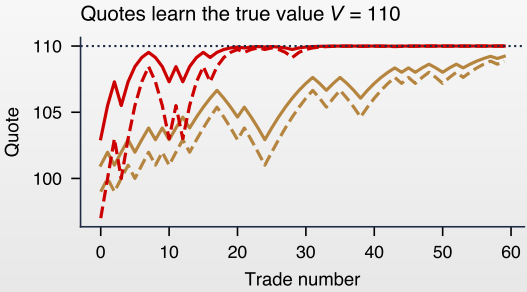
- Value $V \in \{V_L, V_H\}$, $P(V = V_H) = \theta$; each period one trader arrives Glosten & Milgrom (1985)
  - With probability μ informed (buys if $V = V_H$, sells if $V = V_L$), otherwise uninformed, trading for liquidity needs (buys or sells with probability 1/2) Bagehot (1971)
- Competitive, risk-neutral market maker: zero expected profit on each trade, so each quote equals the expected value given the trade it receives
  - $a = \mathbb{E}[V \mid \text{buy}], \qquad b = \mathbb{E}[V \mid \text{sell}]$
  - a: ask, b: bid; a buy signals a high value and a sell a low one, so $a > b$
- Probability of a buy in each state: informed traders always buy when $V = V_H$, the uninformed half of the time
  - $P(\text{buy} \mid V_H) = \mu + \frac{1-\mu}{2}, \qquad P(\text{buy} \mid V_L) = \frac{1-\mu}{2}$
- After each trade, Bayes' rule updates θ: a buy raises it, a sell lowers it

Glosten–Milgrom: the Spread

- Bayes' rule: the probability of the high value after a buy, then the ask as the expected value
  - $P(V_H \mid \text{buy}) = \dfrac{\left(\mu + \frac{1-\mu}{2}\right)\theta}{\left(\mu + \frac{1-\mu}{2}\right)\theta + \frac{1-\mu}{2}(1-\theta)}, \qquad a = V_L + (V_H - V_L)\,P(V_H \mid \text{buy})$
  - Numerator: probability of "high value and a buy"; denominator: total probability of a buy; the bid follows symmetrically from a sell
- At $\theta = 1/2$: $a - b = \mu(V_H - V_L)$
  - The spread grows linearly with the share of informed traders
  - $V_L = 90$, $V_H = 110$: spread 2 for $\mu = 0.1$, 6 for $\mu = 0.3$
- No inventory, no fees: the spread exists purely because of information
- Trade prices form a martingale: the market is efficient in the semi-strong sense (Chapter 2)

Glosten–Milgrom in Action (1/2)





Initial spread ask - bid ($V_L = 90$, $V_H = 110$) — Ask, $\mu = 0.1$ — Bid, $\mu = 0.1$ — Ask, $\mu = 0.3$ — Bid, $\mu = 0.3$

MFM_ch10_information

Glosten–Milgrom in Action (2/2)

- Left: initial spread as a function of μ
- Right: one simulated sequence of 60 trades with true value $V = 110$ (same random draws for both μ)
- On these individual paths the bid exceeds 108 after 7 trades for $\mu = 0.3$, only after 48 for $\mu = 0.1$
- More informed traders: wider spreads, but faster price discovery, a trade-off between liquidity and informational efficiency; empirical counterpart: PIN (probability of informed trading) Easley, Kiefer, O'Hara & Paperman (1996)

MFM_ch10_information

The Kyle Model

- One insider knows $v \sim N(p_0, \sigma_v^2)$ and submits a market order x Kyle (1985)
  - Noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v
  - The market maker sees only the total order flow $y = x + u$ and sets $p = \mathbb{E}[v \mid y]$
- The insider chooses x to maximise $\mathbb{E}[(v - p)\, x]$, knowing that his own order moves the price
- Linear equilibrium: the insider trades in proportion to his advantage, the price moves in proportion to the order flow
  - $x = \beta(v - p_0), \qquad p = p_0 + \lambda y$
  - p_0: price before trading (the prior mean of v); σ_v, σ_u: standard deviations of the value and of noise trading
  - $\beta > 0$: trading intensity of the insider; $\lambda > 0$: **Kyle's lambda**, the price move per share of net order flow

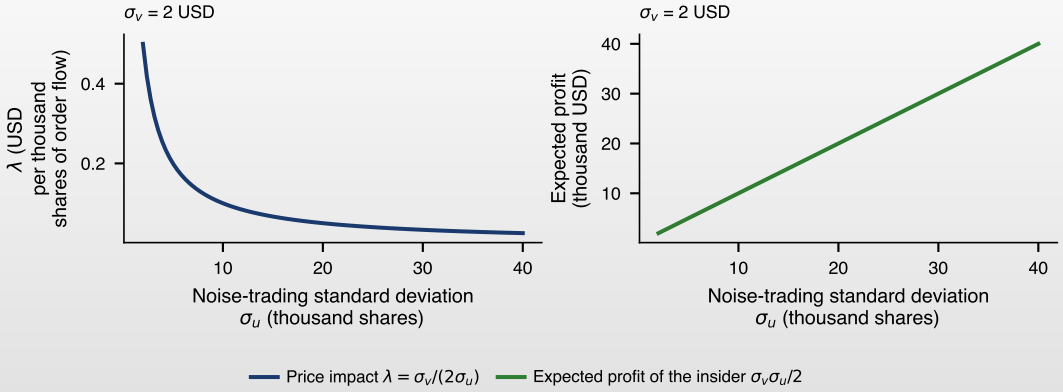
Kyle: Derivation and Equilibrium (1/2)

- **Insider**: takes the pricing rule $p = p_0 + \lambda(x + u)$ as given and maximises his expected profit, knowing v
  - $\mathbb{E}[(v - p)\, x \mid v] = (v - p_0)x - \lambda x^2$ (since $\mathbb{E}[u] = 0$)
  - First term: gain from the information; $-\lambda x^2$: the cost of moving the price against himself
  - First-order condition (derivative in x set to 0): $x = (v - p_0)/(2\lambda)$, so $\beta = 1/(2\lambda)$: he trades less because his order moves the price
- **Market maker**: (v, y) is jointly Normal, so $\mathbb{E}[v \mid y]$ is the linear projection of v on y
  - $\lambda = \mathrm{Cov}(v, y)/\mathrm{Var}(y) = \beta\sigma_v^2/(\beta^2\sigma_v^2 + \sigma_u^2)$
  - The population OLS (ordinary least squares) slope of v on y: how much one share of order flow tells about the value

Kyle: Derivation and Equilibrium (2/2)

- Substituting $\beta = 1/(2\lambda)$ into the slope gives $\beta^2\sigma_v^2 = \sigma_u^2$
- **Equilibrium**
  - $\beta = \sigma_u/\sigma_v, \quad \lambda = \sigma_v/(2\sigma_u), \quad \mathbb{E}[\text{insider profit}] = \sigma_v\sigma_u/2$
  - More noise trading (σ_u up): a smaller λ (deeper market), a larger β and a larger profit
  - More valuable information (σ_v up): a larger λ, the market maker protects himself
- What the price reveals
  - $\mathrm{Var}(x) = \beta^2\sigma_v^2 = \sigma_u^2$: the insider's orders are as variable as the noise, so he hides behind noise traders
  - $\mathrm{Var}(v \mid y) = \sigma_v^2 - \lambda^2\mathrm{Var}(y) = \sigma_v^2/2$: the uncertainty left after trading; the price reveals half of the private information

Kyle's Lambda and the Insider's Profit (1/2)



MFM_ch10_information

Kyle's Lambda and the Insider's Profit (2/2)

- $\sigma_v = 2$ USD; noise trading σ_u from 2 000 to 40 000 shares
- Chart units: λ in USD per thousand shares; $1/\lambda$ is **market depth**
- Reading the chart: λ falls like $1/\sigma_u$, the insider's profit rises linearly in σ_u

MFM_ch10_information

How Should We Interpret PIN?

- PIN: α probability of an information event on a day, μ arrival rate of informed orders, ε_b, ε_s arrival rates of uninformed buys and sells; $\mathrm{PIN} = \alpha\mu/(\alpha\mu + \varepsilon_b + \varepsilon_s)$, PIN $\in [0, 1]$
- Example: $\alpha = 0.5$, $\mu = 2$, $\varepsilon_b = \varepsilon_s = 2$ give $\mathrm{PIN} = 1/(1 + 2 + 2) = 0.20$: one fifth of expected trades are informed
- VPIN (volume-synchronised PIN), order imbalance in volume time Easley, López de Prado & O'Hara (2012), is contested: correctly computed it did not predict the Flash Crash and largely tracks volume and volatility Andersen & Bondarenko (2014)
- PIN describes a model-implied share; it does not identify which particular trade was informed

From Models to Measurement

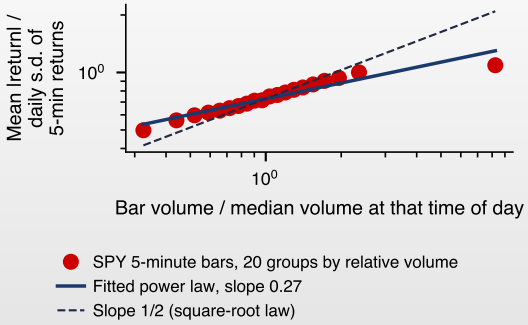
Measure	What it identifies	Data needed
Kyle λ	price change per unit of signed order flow; the Amihud ratio is its daily proxy Hasbrouck (2009)	signed flow, or daily bars
MRR	Madhavan–Richardson–Roomans model: permanent (adverse-selection) and transitory parts of the spread Madhavan, Richardson & Roomans (1997)	signed trades and quotes
Hasbrouck VAR	vector autoregression (VAR): long-run quote response to a trade innovation Hasbrouck (1991)	signed trades and quotes
PIN	$\alpha\mu/(\alpha\mu + \varepsilon_b + \varepsilon_s)$, the expected share of informed trades Easley, Kiefer, O'Hara & Paperman (1996)	daily buy and sell counts
Order-flow imbalance	explains most short-horizon price changes Cont, Kukanov & Stoikov (2014)	order-book events

- The public bars used here do not sign trades, so MRR, the VAR and PIN are not estimated here
- Their equations are in the Appendix: ▸ MRR ▸ PIN ▸ Hasbrouck VAR

Metaorders and the Square-Root Law

- Signs of market orders are autocorrelated for hundreds of trades: large **metaorders** are split into many small trades over hours or days Bouchaud, Gefen, Potters & Wyart (2004) (propagator and Hawkes models: Appendix)
- Empirical law for the peak impact (from the first to the last trade of the metaorder) Tóth et al. (2011), Bouchaud, Bonart, Donier & Gould (2018)
  - $\Delta P \approx Y \sigma_{\text{day}} \sqrt{Q/V_{\text{day}}}$, with ΔP the relative price move caused by the metaorder, Q its size, V_{day} the daily volume, σ_{day} the daily volatility, Y a constant of order 1
- Concave: trading 1% of daily volume moves the price by about $0.1\,\sigma$, not $0.01\,\sigma$: liquidity near the price is scarce and refills slowly (section *The Limit Order Book*)
- Cost: if impact builds as $I(q) = Y\sigma\sqrt{q/V}$ during execution (q: quantity executed so far), the average price paid is $\frac{2}{3} I(Q)$ (before spread and fees)
- Testing it needs the traders' own order records; public bars allow only an indirect look (next slide)

Volume and Price Moves in SPY Bars



- 115 052 five-minute bars in 20 equal-size groups by relative volume (bar volume / median volume at that time of day)
- Mean $|r|$, scaled by the day's volatility, rises with slope 0.27 on log-log axes: concave, but flatter than 1/2
- An association between bars, not a metaorder law
  - news drives both volume and $|r|$ (simultaneity), and opposite trades in a bar net out
- Seminar B6 tries an instrumental variable and questions its exclusion restriction; for execution costs calibrate the metaorder law on your own trades

MFM_ch10_impact

Optimal Execution: the Trade-Off

- Sell X shares within T days, in N steps of length $\tau = T/N$; holdings x_k with $x_{\mathbf{0}} = X$, $x_N = 0$; trades $n_k = x_{k-\mathbf{1}} - x_k$
  Almgren & Chriss (2001)
- Price dynamics: the market price S_k after step k moves with news and is pushed down by our own sales
  - $S_k = S_{k-\mathbf{1}} + \sigma\sqrt{\tau}\,\xi_k - \gamma\, n_k, \qquad \tilde{S}_k = S_{k-\mathbf{1}} - \eta\, n_k/\tau$
  - σ: price volatility (USD per day$^{\mathbf{1/2}}$); ξ_k: news shock, i.i.d., mean 0, variance 1
  - $\gamma \geq 0$: **permanent** impact, lowers all later prices; $\eta > 0$: **temporary** impact, paid only by the trade itself; $\tilde{S}_k$: price we actually receive
- **Trading fast**: high temporary impact, low price risk; **trading slowly**: the opposite
- Objective: a mean-variance trade-off over deterministic schedules
  - min $\mathbb{E}[C] + \lambda \mathrm{Var}[C]$
  - $C = XS_{\mathbf{0}} - \sum_k n_k \tilde{S}_k$: the implementation cost, value at the initial price $S_{\mathbf{0}}$ minus what the sales bring
  - $\lambda \geq 0$: risk aversion (not Kyle's λ); $\lambda = 0$ minimises the expected cost only

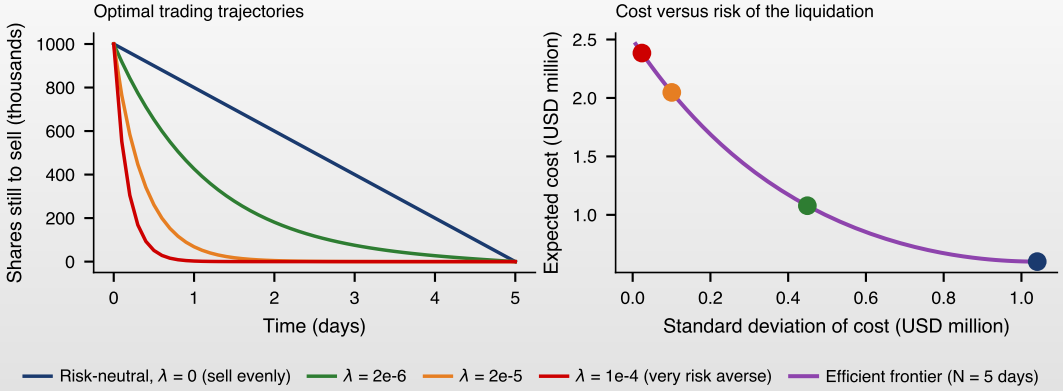
Almgren–Chriss: the Optimal Trajectory (1/2)

- Continuous time: holdings $x(t)$ at time $t \in [0, T]$, selling speed $-\dot{x}$ ($\dot{x} = dx/dt$)
  - Minimise impact cost plus λ times price-risk variance, accumulated over the horizon
  - $\min \int_0^T (\eta\, \dot{x}^2 + \lambda\sigma^2 x^2)\, dt, \qquad x(0) = X,\ x(T) = 0$
  - $\eta\, \dot{x}^2$: selling fast is costly; $\lambda\sigma^2 x^2$: holding a large position is risky
- Solution (derivation: Appendix, seminar A7)
  - $x(t) = X\, \dfrac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \qquad \kappa = \sqrt{\lambda\sigma^2/\eta}$
  - $\sinh z = (e^z - e^{-z})/2$; κ: the urgency of the liquidation, it rises with risk aversion and volatility, falls with impact
  - $\lambda = 0$: $\kappa = 0$ and holdings fall linearly (trade evenly); large λ: most of the order is traded at the start, cutting the exposure to price risk
  - $1/\kappa$ is the characteristic time scale of the liquidation

Almgren–Chriss: the Optimal Trajectory (2/2)

- Discrete time: x_k the holdings after trade k, at time $t_k = k\tau$; $n_k = x_{k-1} - x_k$ the shares sold in step k
  - $x_k = X \dfrac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)}$, $\qquad \frac{2}{\tau^2}(\cosh \kappa\tau - 1) = \frac{\lambda\sigma^2}{\tilde{\eta}}$, $\qquad \tilde{\eta} = \eta - \frac{\gamma\tau}{2} > 0$
  - Same shape as in continuous time; κ now solves the middle equation ($\cosh z = (e^z + e^{-z})/2$), which tends to $\kappa^2 = \lambda\sigma^2/\eta$ as $\tau \to 0$
  - $\tilde{\eta}$: temporary impact corrected for the discrete steps
- Mean and variance of the cost of any schedule
  - $\mathbb{E}[C] = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau}\sum_{k=1}^{N} n_k^2$, $\qquad \mathrm{Var}[C] = \sigma^2\tau\sum_{k=1}^{N} x_k^2$
  - $\frac{1}{2}\gamma X^2$: permanent impact, the same for every schedule; $\sum n_k^2$: penalises large trades; $\sum x_k^2$: penalises holding the position long

Optimal Trajectories and the Efficient Frontier (1/2)



MFM_ch10_execution

Optimal Trajectories and the Efficient Frontier (2/2)

- Numerical example of Almgren & Chriss (2001): $X = 10^6$ shares, $S_0 = 50$ USD, $T = 5$ days, $\sigma = 0.95$ USD per day$^{1/2}$, $\eta = 2.5 \cdot 10^{-6}$, $\gamma = 2.5 \cdot 10^{-7}$
- With $\lambda = 2 \cdot 10^{-6}$ the first day sells 57% of the position; the standard deviation of cost falls from 1.04 (even selling) to 0.45 million USD for 0.48 million USD more expected cost
- The frontier is convex: the first units of risk reduction are cheap, the last ones expensive (full table: Appendix)

MFM_ch10_execution

Execution Benchmarks

- **Implementation shortfall**: paper portfolio at the decision price minus the executed portfolio Perold (1988)
- VWAP: the volume-weighted average price over the period; TWAP: the time-weighted one, matched by the $\lambda = 0$ schedule
- A participation rule (a fixed share of market volume) ignores price: the rule at the heart of the Flash Crash (next section)

High-Frequency Trading: the New Market Makers

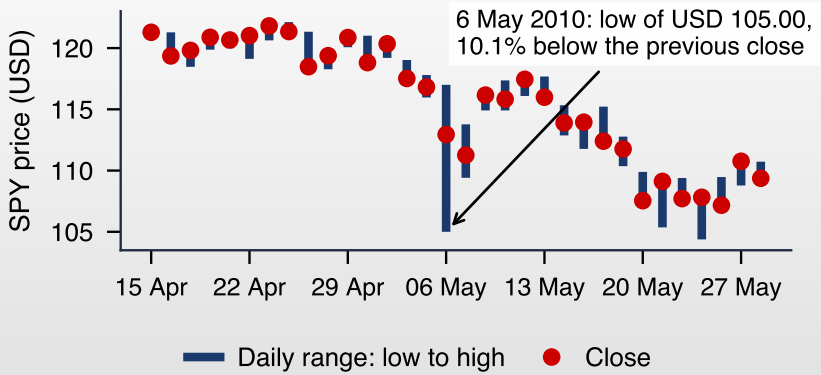


A microwave relay tower (Mojave, California); similar line-of-sight links carry trading signals between Chicago and New Jersey

Photo: Tony Webster (2016); CC BY-SA 2.0; Wikimedia Commons

- HFT (high-frequency trading): algorithms that trade in micro- to milliseconds and end the day flat
  - speed from co-location (servers next to the exchange's matching engine) and microwave links
- Low latency speeds up price discovery and narrows spreads Hasbrouck & Saar (2013)
- One HFT market maker on Chi-X and Euronext Menkveld (2013)
  - mostly passive orders: it earned the spread and lost on positions
- Order flow and latency become sources of information in their own right O'Hara (2015)
- HFT market makers as intermediaries: Jovanovic & Menkveld (2010)
- Inventory skews quotes: a dealer who is long lowers both quotes to sell Avellaneda & Stoikov (2008) (Appendix)
- HFT liquidity is plentiful in calm markets and fragile under stress

The Flash Crash of 6 May 2010



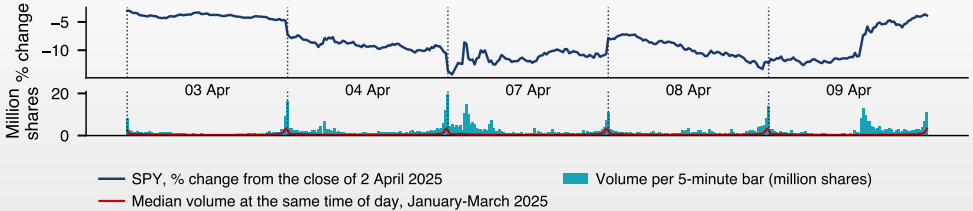
MFM_ch10_stress

Flash Crash: Timeline and Regulatory Responses

- **Daily** SPY range and close, 15 Apr – 28 May 2010 (unadjusted): the bars cannot show the intraday collapse and recovery
- 2:32 p.m.: a mutual fund complex starts selling (report of the Commodity Futures Trading Commission (CFTC) and Securities and Exchange Commission (SEC): CFTC & SEC (2010))
  - 75 000 E-mini S&P 500 futures at 9% of volume, regardless of price
  - 2:45:28 p.m.: a 5-second pause lets buyers return
- SPY low USD 105.00, 10.1% below the previous close
  - previous close 116.82; close only 3.3% lower; volume 3.0 times the recent median
- HFT did not trigger the crash but amplified it Kirilenko, Kyle, Samadi & Tuzun (2017)
  - responses: single-stock circuit breakers (replaced in 2013 by the Limit Up–Limit Down mechanism), rules for erroneous trades (details: ▸ Appendix)

MFM_ch10_stress

A Recent Stress Episode: April 2025



- SPY 5-minute bars, 3–9 April 2025; bottom: bar volume against the Jan–Mar 2025 median at the same time of day
- 7 April: range 8.6%, realised volatility 6.7% (median 0.61%)
  - one bar +4.28% and one ten minutes later −2.92%
  - 9 April: +10.0% open to close; volume 4.6 times usual
- Bars document sharp moves and heavy trading; they cannot establish book depth, price continuity or the absence of halts

MFM_ch10_stress

Crypto Venues and Arbitrage

- 24/7 trading with no auctions; the same asset trades on many order books of centralised exchanges (CEX) at once, and in pools of decentralised exchanges (DEX) priced by a formula (automated market maker, AMM, Chapter 16)
  - The same Bitcoin can trade at different prices on different exchanges Makarov & Schoar (2020)
- Price gaps are much larger **across countries** than within a country
  - Capital controls limit the movement of arbitrage capital
- A common component of signed volume across exchanges explains 80% of Bitcoin returns: price discovery is shared across venues Makarov & Schoar (2020)
- Crypto liquidity can be forecast from market data Agakishiev, Härdle & Pele (2025)

Price Discovery across Venues: Information Shares (1/2)

- Two log prices of one asset on two venues, $y_t = (p_{1t}, p_{2t})'$: each is I(1) (a random walk-like series), their difference is stationary
  - Cointegrated with vector $\beta = (1, -1)'$: $\beta' y_t = p_{1t} - p_{2t}$ is the price gap between venues
- VECM (vector error-correction model) Johansen (1991): each price reacts to yesterday's gap and to past changes
  - $\Delta y_t = \alpha\, \beta' y_{t-1} + \sum_{i=1}^{p} \Gamma_i\, \Delta y_{t-i} + e_t, \qquad \mathrm{Var}(e_t) = \Omega$
  - $\alpha = (\alpha_1, \alpha_2)'$: adjustment coefficients, the speed at which each price closes the gap; Γ_i: 2×2 matrices of short-run dynamics, p lags
  - e_t: innovations of the two prices, with covariance matrix Ω
- Common efficient price: the stochastic trend shared by both prices
  - Its innovation is $\psi' e_t$, with loadings $\psi' \propto \alpha'_\perp$, where $\alpha_\perp$ is orthogonal to α ($\alpha'_\perp \alpha = 0$); $\propto$: proportional to

Price Discovery across Venues: Information Shares (2/2)

- **Hasbrouck information share** Hasbrouck (1995): venue j's share of the variance of the common-trend innovation
  - $IS_j = \dfrac{([\psi' F]_j)^2}{\psi' \Omega \psi} \in [0,1], \qquad IS_1 + IS_2 = 1$
  - F: Cholesky factor of Ω (lower triangular, $FF' = \Omega$); $[\cdot]_j$: the j-th element
  - Depends on the ordering when Ω is not diagonal: report the bounds over both orderings
- **Gonzalo–Granger component share** Gonzalo & Granger (1995): $\alpha_\perp$ normalised to sum 1
  - For two venues $CS_1 = \alpha_2/(\alpha_2 - \alpha_1)$; in $[0,1]$ only if the two adjustment coefficients have opposite signs
- Reading: a price that does **not** adjust ($\alpha_j = 0$, weak exogeneity) leads price discovery, share 1
  - Inference: a Wald test of $\alpha_j = 0$; non-rejection does not prove $\alpha_j = 0$

Price Discovery on the BVB: the BET ETF and the BET-TR Index (1/2)

- Daily log closes of the Patria-TVBETETF (it reinvests dividends) and the BET-TR index (BET Total Return, dividends reinvested), 495 common trading days, September 2024 – September 2026
- The basis (log ETF minus log index) has standard deviation 0.84%; BET-TR is computed from its constituents' prices, not traded on a venue
- **Johansen trace test**: tests the number r of cointegrating vectors, step by step from $r = 0$
  - 49.6 for $r = 0$ (5% critical value 15.49): rejected; 0.10 for $r \leq 1$ (3.84): not rejected
  - Hence one cointegrating vector; VECM with 1 lag, chosen by BIC (Bayesian information criterion)

Price Discovery on the BVB: the BET ETF and the BET-TR Index (2/2)

- Adjustment: ETF $\alpha = -0.12$ ($t = -2.11$); index $\alpha = -0.03$ ($t = -0.63$)
  - The index adjustment is not distinguishable from zero, the ETF's is: suggestive evidence that the ETF adjusts toward the index, with wide bootstrap intervals (seminar B8)
- Information share of the index: between 0.25 and 0.98; residual correlation 0.93. Component share of the index 1.39, outside $[0, 1]$ because both α are negative
- Lesson: with daily closes the innovations are almost collinear, so the bounds are wide; price discovery needs intraday data (seminar B8)

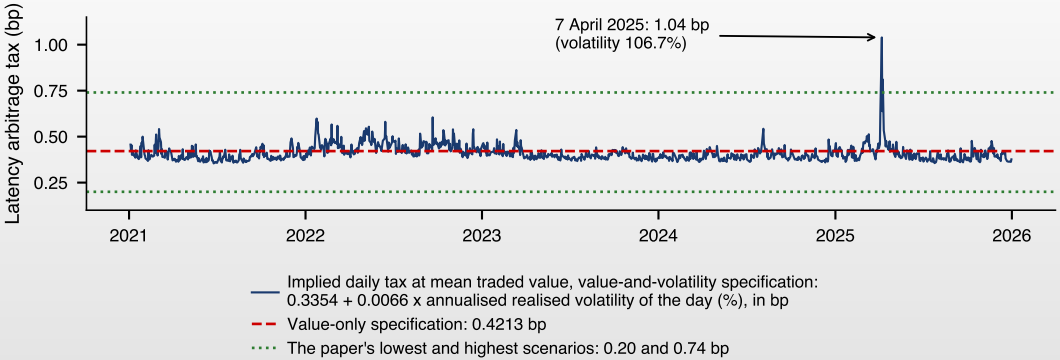
Case Study: the Speed Race and the Question

- **Latency arbitrage** (sniping): trading against quotes made stale by a public price move before their provider can cancel them Budish, Cramton & Shim (2015)
  - In a continuous book the fastest wins; liquidity providers pay through wider spreads and thinner books; proposal: **frequent batch auctions**, e.g. every second, moving competition from speed to price
- **Question** Aquilina, Budish & O'Neill (2022): how much does the race cost a whole stock market?
- Why it was hard: order-book data record only the winners; a failed order or a cancel that arrives too late leaves the book unchanged
  - **Message data**: every message to and from the exchange, including "too late" replies, so both winners and losers are seen
- Link to the section *The Bid-Ask Spread*: is sniping on **public** news a cost of liquidity next to **private**-information adverse selection (section *Information and Prices*)?

Case Study: Measurement and Three Central Results

- **Measurement chain**: London Stock Exchange messages, FTSE 350 (the 350 largest London-listed companies), 43 days of 2015, about 2.2 billion messages (definitions: Appendix)
  - Race: 2+ participants, same stock, price and side, within the information horizon (at most 500 μs, microseconds), at least one success and one failure; race profit marked to the mid 10 s later
  - **Latency arbitrage tax**: daily race profits / daily traded value, in bp
- **1. Frequent and fast**: 537 races a day per FTSE 100 stock, about one a minute, decided by 5–10 μs; 22% of FTSE 100 traded value is in races
- **2. Concentrated**: the top six firms win 82% of races (a count of races); a race is mostly a transfer from slower liquidity providers to a few fast takers
- **3. Small prizes, large sums**: tax 0.419 bp of all traded value; eliminating races would cut the cost of liquidity by about 17%; USD 4.8 billion a year in global equities

Case Study: the Implied Daily Tax on SPY (1/2)



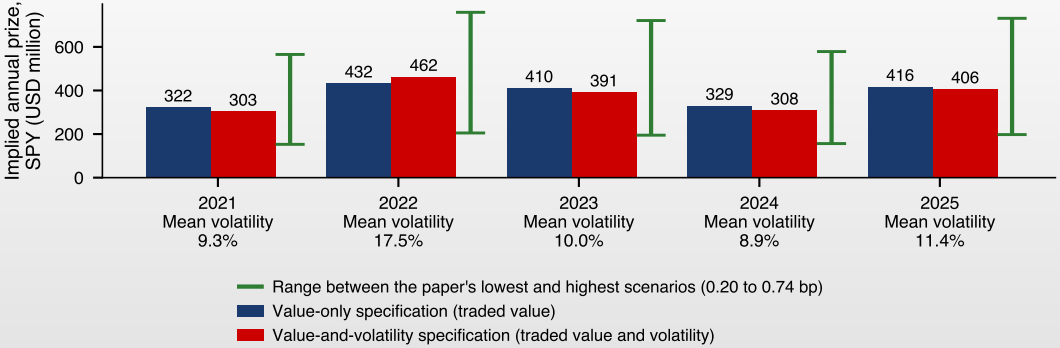
MFM_ch10_race

Case Study: the Implied Daily Tax on SPY (2/2)

- Implied tax $0.3354 + 0.0066\,\sigma_t$ bp, from the specification with traded value and volatility, at mean traded value
- σ_t: annualised realised volatility (%) from SPY 5-minute bars, 1 244 full days of 2021–2025
- The median tax is 0.40 bp, close to the 0.419 bp measured in London
- On 30% of days the tax exceeds 0.4213 bp, the value of the specification with traded value only
- The peak, 1.04 bp, is on 7 April 2025 ($\sigma_t = 106.7\%$); only 2 days exceed the authors' highest scenario (0.74 bp)
- The volatility term matters only on stress days: most of the prize scales with traded value

MFM_ch10_race

Case Study: the Implied Annual Prize for SPY (1/2)



MFM_ch10_race

Case Study: the Implied Annual Prize for SPY (2/2)

- Annual prize $\sum_t \Pi_t$, USD million; Π_t: implied race profit on day t (tax $\times$ traded value)
- Bars: the specification with traded value only (0.4213 bp) and the one with value and volatility; whiskers: the authors' scenarios, 0.20–0.74 bp
- 2025: 416 million (value only) and 406 million (value and volatility)
- The volatility specification gives more only when the year's mean volatility exceeds $(0.4213 - 0.3354)/0.0066 = 13.0\%$: in 2022 (17.5%), not in the other years (8.9–11.4%)
- The scenarios differ by a factor of 3.7, the specifications by at most 7%: the tax level is the main uncertainty

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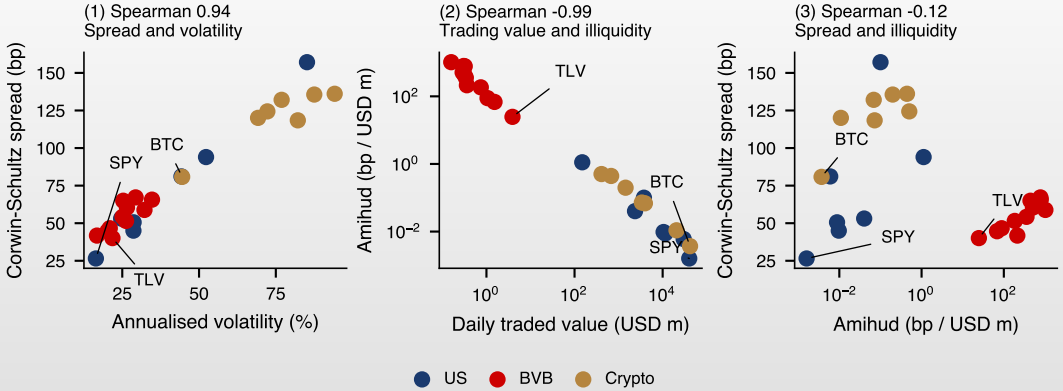
Case Study: What the Transfer Assumes

- **Method**: coefficient-transfer extrapolation: the coefficients estimated in London are applied to SPY's traded value and volatility; races are not re-estimated on SPY (no US message data)
  - It assumes the same race technology and the same tax per unit of value; volatility here is from 5-minute returns, the authors use 1-minute returns
- **Limits stated by the authors**: one exchange, nine weeks; market structure (fragmentation, ETFs) and other asset classes ignored; long races and silent losers missed
  - For scale: the futures–SPY arbitrage of Budish, Cramton & Shim (2015) was about USD 75 million a year; the implied SPY prize is four to six times larger
- **Open extension**: does SPY's daily EDGE spread co-move with the implied tax? Both depend on volatility, so this is a diagnostic; identification needs a message-based race measure and a matched effective-spread benchmark

An Open Question: Do Price-Based Proxies Measure Trading Costs?

- Where trades and quotes are unavailable (BVB history, many crypto venues), costs are estimated from daily OHLC bars
  - Jahan-Parvar & Zikes (2023): low-frequency estimators track effective spreads only when the spread is large relative to volatility; otherwise volatility dominates
  - Ardia, Guidotti & Kroencke (2024): EDGE, an estimator from open, high, low and close prices, unbiased and efficient under their assumptions
- Formal question: $\hat{S}_i = S_i + b(\sigma_i) + \varepsilon_i$ across assets i
  - $\hat{S}_i$: proxy estimate for asset i; S_i: its true spread; $b(\sigma_i)$: bias as a function of the asset's volatility σ_i; ε_i: estimation noise
  - H_0: $b \equiv 0$ (the proxy ranks assets by cost) against H_1: b increasing in σ_i; $\equiv$: identically, for every σ_i
  - Identification needs a cost benchmark measured independently of the proxy: effective spreads from matched trades and quotes; Amihud measures price impact, not the spread, and itself contains $|r|$
- Open: where direct cost data are scarce, is the spread-to-volatility ratio high enough for reliable estimation? Thin trading alone does not determine that ratio

Mini-Case: Cost or Volatility?



MFM_ch10_seminar

Testing the AI Claim with Spearman Rank Correlations

- A typical claim of an artificial intelligence (AI) assistant: "the estimates are similar, so BVB blue chips are about as liquid as US large caps"; test on the 25 assets, Sep 2024 – Sep 2026 (Spearman rank correlation $\rho \in [-1, 1]$, p: its p-value)
- Corwin–Schultz with volatility $\rho = 0.94$ ($p < 0.001$), with Amihud $\rho = -0.12$ ($p = 0.58$)
  - Difference between the two Spearman correlations 1.06, 95% CI $[0.58, 1.44]$ over assets, $[0.71, 1.43]$ within markets
  - Partial Corwin–Schultz–Amihud ρ controlling for volatility 0.27
- Daily Corwin–Schultz ranks track volatility, as Jahan-Parvar & Zikes (2023) predict; a diagnostic, not a test of H_0: Amihud is not an independent spread benchmark

MFM_ch10_seminar

The Discovery Loop and a Project Seed

- **Loop**, with an AI assistant (prompts: Appendix)
  - 1. Literature: collect validation studies and check every DOI (digital object identifier) on Crossref; 2. rival hypotheses, each with a cross-sectional prediction
  - 3. Replicate the chapter's numbers first (medians US 53, BVB 54, crypto 124 bp); 4. a pre-registered robustness plan; 5. ask the assistant to attack the benchmark; 6. report everything tried
- **Seed**: which estimator from daily bars, if any, measures the level of trading costs on the BVB and for crypto-assets?
- Deliverables
  - A dated pre-registration; EDGE implemented after Ardia, Guidotti & Kroencke (2024) and validated on simulated prices with a known spread; code and a two-page report with every specification tried
  - Declare AI use in `AI_USE.md` and the errors caught in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

Which Liquidity Measure for Which Data?

Available data	Suggested measure
Trades and quotes with timestamps	Effective and realised spread; MRR by the generalised method of moments (GMM); Hasbrouck VAR; PIN
Prices of one asset on several venues	VECM; information-share bounds and component shares
Intraday bars (1–5 minutes)	Corwin–Schultz, Abdi–Ranaldo or EDGE per day (EDGE is the closest to one tick for SPY); intraday Amihud
Daily prices and volumes	Amihud ratio (price impact) across assets; EDGE for spread levels only if validated against effective spreads; high-low estimators mainly for changes over time
Own executions	Implementation shortfall; calibrate the square-root law
Theory, what-if analysis	Simulated order books; Glosten–Milgrom, Kyle, Almgren–Chriss

Check Your Understanding, and the Road to Continuous Time

Can you answer these?

- ⊡ Why can a positive Roll covariance occur even when the Roll model is true?
- ⊡ Why do daily high-low estimators rank BVB stocks next to US large caps?
- ⊡ In Kyle's model, what happens to λ and to the insider's profit when noise trading doubles?
- ⊡ Why does a risk-averse seller front-load?

Chapter 11: continuous time

- ⊡ From trades and order books back to continuous-time price models: Brownian motion and Itô calculus
- ⊡ At high frequency the bid-ask bounce is microstructure noise that contaminates realised volatility (Chapter 9)

Order Types and the Simulated Book

- Conditions on execution and lifetime
  - IOC (immediate-or-cancel): executes at once as far as possible, the rest is cancelled; FOK (fill-or-kill): executes in full at once or is cancelled; GTC (good-till-cancelled): waits in the book until cancelled
- **Stop** order: becomes a market order once a trigger price trades; **iceberg**: displays only part of its size, the rest waits hidden
- A marketable limit order (a buy limit at or above the best ask) consumes liquidity like a market order, with a price cap
- **Zero-intelligence** simulation Farmer, Patelli & Zovko (2005)
  - Limit orders: rate α per price level, uniform up to 40 ticks from the opposite best quote; market orders: rate μ per side; cancellations: rate δ per standing order
  - Here $\alpha = 0.1$, $\mu = 0.5$, $\delta = 0.02$; 300 000 events, unit orders; simulation rates, unrelated to the PIN parameters with the same letters
  - Such a model already predicts much of the cross-section of spreads on the London Stock Exchange

Back

Statistical Properties of the Roll Estimator

- Probability of a wrong sign under the Gaussian approximation Harris (1990) (bias and variance of $\hat{\gamma}_1$: slide *The Roll Estimator in Finite Samples*)
  - $P(\hat{\gamma}_1 > 0) \approx \Phi\left(\sqrt{T}\,\gamma_1 / \sqrt{\gamma_0^2 + 3\gamma_1^2}\right)$ ignoring the demeaning bias; it reduces to $\Phi(\sqrt{T}\,\gamma_1/\gamma_0)$ of the main text when $c^2 \ll \sigma^2$
- Delta method (a first-order Taylor expansion of $\hat{s} = 2\sqrt{-\hat{\gamma}_1}$ around γ_1): the standard error
  - $\mathrm{se}(\hat{s}) \approx \sqrt{(\gamma_0^2 + 3\gamma_1^2)/T}\,/\,c$
  - Large when c is small: the smaller the spread, the less precisely it is estimated; $\hat{s}$ is also biased (Jensen's inequality, truncation at 0)
- With daily data the bounce (a few bp) is buried under the daily volatility (103 bp for SPY over the last two years), hence estimators based on the daily high and low

Back

A Structural Model of Trading Costs: MRR (1/2)

- Roll assumes i.i.d. trade signs; real signs are persistent (a buy tends to follow a buy)
  - First-order Markov signs: $\text{Corr}(q_t, q_{t-k}) = \rho^k$, $\rho \in [0, 1)$ the persistence; Corr: correlation
  - Signs independent of ε and no information: $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2(1-\rho)^2$, closer to 0 than $-c^2$, so Roll underestimates c
- MRR model Madhavan, Richardson & Roomans (1997): the trade price is the efficient price plus a transitory cost; the efficient price moves with the **surprise** in the order flow
  - $p_t = m_t + \phi\, q_t + \xi_t, \qquad m_t = m_{t-1} + \theta\,(q_t - \mathbb{E}[q_t \mid q_{t-1}]) + \varepsilon_t$
  - $\theta \geq 0$: permanent (adverse-selection) impact of the surprise $q_t - \mathbb{E}[q_t \mid q_{t-1}] = q_t - \rho q_{t-1}$; $\phi \geq 0$: transitory cost (processing, inventory)
  - q_t a first-order Markov chain; public news ε_t and rounding noise ξ_t i.i.d., mean zero, independent of each other and of q

Back

A Structural Model of Trading Costs: MRR (2/2)

- Estimation by GMM: choose (θ, ϕ, ρ, a) so that the sample versions of four moments are zero
  - Residual: $u_t = \Delta p_t - (\phi + \theta) q_t + (\phi + \rho\theta) q_{t-1}$, the price change not explained by the trades; a: its drift (constant)
  - $\mathbb{E}[q_t q_{t-1} - \rho q_{t-1}^2] = 0, \quad \mathbb{E}[u_t - a] = 0, \quad \mathbb{E}[(u_t - a) q_t] = 0, \quad \mathbb{E}[(u_t - a) q_{t-1}] = 0$
  - First: defines ρ as the autocorrelation of signs; the others: the residual has mean a and is uncorrelated with current and past signs
- Outputs
  - Implied effective spread $2(\phi + \theta)$; adverse-selection share $\theta/(\phi + \theta) \in [0, 1]$: 0 means pure processing and inventory costs, 1 pure information
- Needs signed trades (trade-and-quote data); the public 5-minute bars do not identify q_t, so MRR is not estimated here

Back

Corwin–Schultz and EDGE in Practice

- Corwin–Schultz on daily data: the derivation assumes no price move overnight; the paper shifts H_{t+1}, L_{t+1} down by $L_{t+1} - C_t$ if $L_{t+1} > C_t$, up by $C_t - H_{t+1}$ if $H_{t+1} < C_t$ (C_t the close) Corwin & Schultz (2012)
- Within one session of 5-minute bars no overnight adjustment is needed; negative two-bar estimates are set to 0
- EDGE over windows: the authors recommend setting each negative window estimate to zero and then averaging, since more negative estimates go with larger spreads Ardia, Guidotti & Kroencke (2024)

Back

The EDGE Moment Conditions

- Log prices of bar t: open o_t, high h_t, low l_t, close c_t; mid-range $\eta_t = (h_t + l_t)/2$ as for Abdi–Ranaldo Ardia, Guidotti & Kroencke (2024)
  - Five log-returns around the open and the previous close: $r_{1t} = \eta_t - o_t$, $r_{2t} = o_t - \eta_{t-1}$, $r_{3t} = \eta_t - c_{t-1}$, $r_{4t} = c_{t-1} - \eta_{t-1}$, $r_{5t} = o_t - c_{t-1}$
- Two moment series; their sample means are the estimates $\hat{e}_1$, $\hat{e}_2$ of S^2, their sample variances v_1, v_2
  - $x_{1t} = -\dfrac{4}{p_o}\, d_{1t}\, r_{2t} - \dfrac{4}{p_c}\, d_{3t}\, r_{4t}, \qquad x_{2t} = -\dfrac{4}{p_o}\, d_{1t}\, r_{5t} - \dfrac{4}{p_c}\, d_{5t}\, r_{4t}$
  - $d_{kt} = r_{kt} - \tau_t\, \bar{r}_k/\bar{\tau}$: r_{kt} centred on the bars that moved; $\tau_t = 0$ only for a flat bar ($h_t = l_t = c_{t-1}$)
  - p_o: share of bars with $o_t \neq h_t$ plus share with $o_t \neq l_t$; p_c: the same for c_{t-1} against h_{t-1}, l_{t-1}; both equal 2 when the open and close never touch an extreme
- Why it works: the open and the previous close are trades at the bid or the ask, so their distances to the surrounding mid-ranges have opposite signs on average, as in Roll; $-4/p$ turns this negative product into S^2

Back

Estimating PIN, and Why It Is Fragile (1/2)

- EKOP model (initials of Easley, Kiefer, O'Hara and Paperman) Easley, Kiefer, O'Hara & Paperman (1996): each day an information event occurs with probability α (bad news with probability δ); given the day's information state, daily buy and sell counts B and S are independent Poisson
  - Rates (expected orders per day): uninformed ε_b, ε_s; informed μ added to the buy side on good-news days, to the sell side on bad-news days
- Likelihood of one day: a mixture over the three states (no news, bad news, good news)
  - $L = (1-\alpha)\,\pi(B;\varepsilon_b)\pi(S;\varepsilon_s) + \alpha\delta\,\pi(B;\varepsilon_b)\pi(S;\mu+\varepsilon_s) + \alpha(1-\delta)\,\pi(B;\mu+\varepsilon_b)\pi(S;\varepsilon_s)$
  - $\pi(n;\lambda) = e^{-\lambda}\lambda^n/n!$: Poisson probability of n orders at rate λ; each term: probability of the state times that of the observed (B, S)
  - The sample likelihood is the product of L over days, maximised in $(\alpha, \delta, \mu, \varepsilon_b, \varepsilon_s)$

Back

Estimating PIN, and Why It Is Fragile (2/2)

- Probability of informed trading: expected informed orders over expected total orders
  - $\mathrm{PIN} = \dfrac{\alpha\mu}{\alpha\mu + \varepsilon_b + \varepsilon_s} \in [0, 1]$
  - PIN $= 0.2$: one order in five is expected to come from an informed trader
- Numerics: for liquid stocks ($B \approx 10^4$) $e^{-\varepsilon}\varepsilon^B/B!$ under- or overflows; factor out common terms and use log-sum-exp Lin & Ke (2011)
  - Without it, estimates pile up at boundary values; the likelihood is also multimodal: use many starting values (seminar A9–A10)

Back

The Hasbrouck Trade–Quote VAR

- VAR in the mid-quote return r_t and the signed trade x_t Hasbrouck (1991)
  - $r_t = \sum_{i\geq 1} a_i r_{t-i} + \sum_{i\geq 0} b_i x_{t-i} + \varepsilon_{1t}, \qquad x_t = \sum_{i\geq 1} c_i r_{t-i} + \sum_{i\geq 1} d_i x_{t-i} + \varepsilon_{2t}$
  - t: trade time; $x_t = q_t$ (± 1) or signed volume; a_i, b_i, c_i, d_i: lag coefficients; ε_{1t}: quote shock (public news), ε_{2t}: trade shock (the unexpected part of the order flow)
- Identification: the trade comes first within the interval (b_0 enters, the reverse contemporaneous term does not): a recursive (Cholesky) ordering
- **Permanent impact**: the long-run cumulative response of the quote to a trade shock, $\lim_{h\to\infty} \sum_{k=0}^{h} \partial r_{t+k}/\partial \varepsilon_{2t}$ (the total change in the quote over all later steps k caused by a unit trade shock): the information content of trades
- Variance decomposition of the efficient price: the share explained by trades generalises the adverse-selection share of MRR
- Like MRR, it needs signed trades; in Kyle's model the permanent impact is exactly λ

Back

Order Flow Is Long-Memory and Self-Exciting (1/2)

- Signs of market orders are autocorrelated with a power-law decay: metaorders are split Bouchaud, Gefen, Potters & Wyart (2004)
  - $\mathrm{Corr}(\varepsilon_t, \varepsilon_{t+\ell}) \sim \ell^{-g}, \qquad 0 < g < 1$
  - Here $\varepsilon_t = \pm 1$: sign of the t-th market order; ℓ: lag in number of orders; $\sim$: behaves like, for large ℓ
  - $g < 1$: the correlations decay so slowly that their sum diverges (long memory)
- Yet prices are close to a random walk: the propagator model reconciles the two
  - $p_t = \sum_{s<t} G(t-s)\,\varepsilon_s + \text{noise}$
  - $G(\tau)$: the impact of an order τ steps later; a decaying kernel G offsets the persistent signs

Back

Order Flow Is Long-Memory and Self-Exciting (2/2)

- Arrival times cluster: a Hawkes process has an intensity that jumps after each event Bacry, Mastromatteo & Muzy (2015)
  - $\lambda(t) = \mu + \int_{-\infty}^{t} \phi(t-s)\, dN_s = \mu + \sum_{t_i < t} \phi(t - t_i)$
  - $\lambda(t)$: expected number of events per unit time at t; $\mu > 0$: baseline rate; N_s: number of events up to s, occurring at times t_i
  - $\phi \geq 0$: excitation kernel, how much a past event raises today's rate
- Estimated by maximum likelihood; branching ratio $\|\phi\|_1 = \int_0^\infty \phi(\tau)\, d\tau$
  - The share of events triggered by earlier events (endogeneity); stationary if < 1; close to 1: the market mostly reacts to itself
- Links: durations between exceedances of the value at risk (VaR, not the VAR above) and their clustering (Chapter 8), intraday volatility (Chapter 9), the concave impact (section *Price Impact and Optimal Execution*)

Back

Deriving the Almgren–Chriss Trajectory

- Continuous time, holdings $x(t)$, trading rate $-\dot{x}$: temporary impact cost $\eta \int_0^T \dot{x}^2\, dt$, price-risk variance $\sigma^2 \int_0^T x^2\, dt$
  - Minimise $\int_0^T (\eta\, \dot{x}^2 + \lambda\sigma^2 x^2)\, dt$ subject to $x(0) = X$, $x(T) = 0$ (the permanent impact $\frac{1}{2}\gamma X^2$ does not depend on the path)
- Euler–Lagrange equation (the first-order condition of a calculus-of-variations problem) for the integrand $F(x, \dot{x}) = \eta\dot{x}^2 + \lambda\sigma^2 x^2$: $\frac{d}{dt}\, \partial F/\partial \dot{x} = \partial F/\partial x \Rightarrow \ddot{x} = \kappa^2 x$, $\kappa^2 = \lambda\sigma^2/\eta$
- General solution $Ae^{\kappa t} + Be^{-\kappa t}$, with constants A, B; the boundary conditions give $x(t) = X \sinh(\kappa(T - t))/\sinh(\kappa T)$
- In discrete time κ^2 becomes $\frac{2}{\tau^2}(\cosh \kappa\tau - 1) = \lambda\sigma^2/\tilde{\eta}$ (section *Price Impact and Optimal Execution*); a resilient book with transient impact changes the shape Obizhaeva & Wang (2013)

Back

Almgren–Chriss, the Numerical Table

Strategy (5 daily steps)	Expected cost	Standard deviation
Even selling, $\lambda = 0$	0.60 million USD	1.04 million USD
$\lambda = 2 \cdot 10^{-6}$	1.08	0.45
$\lambda = 2 \cdot 10^{-5}$	2.05	0.10
$\lambda = 10^{-4}$	2.38	0.02

- With $\lambda = 2 \cdot 10^{-6}$, 428 599 shares remain after the first day
- Same parameters as the main example; each row solves the discrete-time κ equation for its λ

Back

Optimal Market Making: Avellaneda–Stoikov (1/2)

- A dealer quotes a bid and an ask around the mid-price until the horizon T Avellaneda & Stoikov (2008)
  - s_t: mid-price, a Brownian motion with volatility σ; q: the dealer's inventory (shares held, negative if short)
  - He maximises expected exponential (CARA, constant absolute risk aversion) utility of terminal wealth, risk aversion $\gamma > 0$ (not the Almgren–Chriss permanent impact γ)
- Order flow: a quote at distance δ from the mid is hit by market orders with intensity
  - $\Lambda(\delta) = Ae^{-k\delta}$
  - $A > 0$: rate of execution at the mid; $k > 0$: how fast the execution rate falls as the quote moves away from the mid

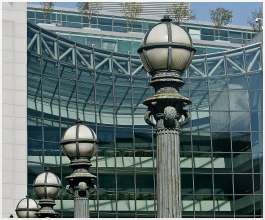
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Optimal Market Making: Avellaneda–Stoikov (2/2)

- Reservation (indifference) price: the value of the asset to this dealer, given his inventory
  - $r = s - q\gamma\sigma^2(T-t)$
  - $T-t$: time left; a long dealer ($q > 0$) values the asset below the mid, and more so with more risk aversion, volatility or time left
- Approximate optimal total spread around r (from an expansion of the value function, not the exact solution)
  - $\delta^a + \delta^b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\left(1+\frac{\gamma}{k}\right)$
  - δ^a, δ^b: distances of the ask and the bid from r; first term: inventory risk; second term: reflects how fast the probability of execution falls with distance from the mid
- Inventory skews both quotes: this formalises the inventory component of the spread (section *The Bid-Ask Spread*)

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Flash Crash: Details from the CFTC–SEC Report



Headquarters of the US Securities and Exchange Commission, Washington

Photo: David (dbking) (2008); CC BY 2.0; Wikimedia Commons

- The CFTC–SEC staff report CFTC & SEC (2010)
- The 2:32 p.m. sell order (main slide) was worth about USD 4.1 billion
  - The algorithm took 9% of the previous minute's volume "without regard to price or time"
- By 2:45:28 p.m. buy-side depth in the E-Mini had fallen below 1% of its morning level
- Over 20 000 trades in more than 300 securities were executed more than 60% away from their earlier prices, some at a penny
- The single-stock circuit breakers of the main slide pause trading for 5 minutes after a 10% move in 5 minutes CFTC & SEC (2010)

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The Bucharest Stock Exchange



Offices of the Bucharest Stock Exchange

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- An electronic limit order book, as in the rest of this chapter BVB (2026)
- Emerging-market status (FTSE Russell) since September 2020; Banca Transilvania and Nuclearelectrica were the first stocks in FTSE emerging-market indices
- A thin market: herding and bubbles in illiquid markets were studied on the BVB by Pele, Mazurencu-Marinescu & Nijkamp (2013)
  - Bubbles and crashes are the subject of Chapter 17

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Case Study: Data and Method

- **Data**: London Stock Exchange message data for all FTSE 350 stocks, 17 August – 16 October 2015, 43 trading days, about 2.2 billion messages, obtained by the British regulator
  - Time stamps accurate to the microsecond (μs), taken where a message reaches the exchange's network, before any exchange processing
- **Race**: 2+ participants on the same stock, price and side; 1+ take and 1+ cancel attempts, or 2+ takes; at least one success and one failure
  - "Same time": within the **information horizon**, the first message's observed processing time plus the fastest reaction time seen (29 μs), capped at 500 μs
- **Race profit**: the signed gap between the race price and the mid-price 10 s later, times the displayed depth
- **Latency arbitrage tax**: daily race profits / daily traded value, in bp; if races vanished, the cost of liquidity would fall by the non-race tax / non-race effective half-spread; averaged over dates, the cut is about 17%
- **Extrapolation**: OLS of daily race profits on traded value V_t, and on V_t and 1-minute realised volatility σ_t times mean traded value $\bar{V}$, without a constant

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Case Study: Further Results

- **Speed**: the winner beats the first loser by 5–10 μs (mode), 46 μs (median); 22.15% of FTSE 100 traded value is in races
- **Concentration**: by volume, the top six firms take about 80% of the liquidity taken in races, but provide only 42% of it; the 82% of the main slide counts races won
- **Prize per race**: per FTSE 100 race, 0.48 ticks (1.20 bp) per share and £1.95 in total; $R^2 = 0.811$ of daily race profits on traded value
  - The USD 4.8 billion a year of the main slide (global equities, 2018) ranges from 2.3 to 8.4 across race definitions
- **Cost of liquidity**: in the full sample, race price impact is 30.58% of all price impact and 32.82% of the effective spread; FTSE 100 realised spread -1.93 bp in races, $+0.15$ bp outside (one-way spreads, half the measures of the section *The Bid-Ask Spread*)
  - This decomposition shows that sniping belongs next to adverse selection in the cost of liquidity

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The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (a large language model, LLM, such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: Semantic Scholar, Elicit, Google Scholar
- **1. Literature**: collect validation studies, then check every DOI on Crossref (the title must match)
  - `Find studies that validate Roll, Corwin-Schultz, Abdi-Ranaldo and EDGE against effective spreads from trades and quotes; report the benchmark, the correlation in levels and in changes, the markets, and the DOI.`
- **2. Competing hypotheses**, each with a cross-sectional prediction
  - `Give rival explanations for why Corwin-Schultz ranks BVB blue chips next to US large caps, and for each the predicted Spearman correlation of the estimate with volatility, with the Amihud ratio and with traded value.`
- **3. Data and code**: replicate the chapter's numbers first
  - `Write Python that loads the daily OHLC prices and volumes of the 25 assets of this chapter and computes Corwin-Schultz, with the overnight adjustment and negative two-day estimates set to zero.` Target medians: US 53, BVB 54, crypto 124 bp

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The Discovery Loop, Reusable Prompts (2/2)

- **4. Robustness**, pre-registered
  - `Write a pre-registered plan: Spearman and partial Spearman correlations, bootstrap over assets, leave-one-market-out (US, BVB, crypto), a second window 2022-2024, and EDGE and Abdi-Ranaldo in place of Corwin-Schultz.`
- **5. Critique**: ask the assistant to attack the benchmark itself
  - `Act as a referee: argue that the Amihud ratio is not a cost measure either (it scales with volume and with volatility); which benchmark would identify the spread, and what would it cost to obtain?`
- **6. Report**: every estimator and window, bootstrap intervals, the benchmark's limits, code and the AI log

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What the Researcher Must Check

- **Tick-size benchmark**: no **quoted** spread is below one tick (effective spreads can be); compare every estimate with $0.01/P$, one tick over the price P (0.21 bp for SPY) as a plausibility check, not a validation
- **Volatility contamination**: test: partial correlation with the benchmark controlling for volatility, and the 5-minute versus daily comparison in the section *The Bid-Ask Spread*
- **Truncation**: negative estimates set to zero bias the mean upwards; report the share truncated and a median
- **Units and venues**: lei converted at the BNR rate; crypto volume summed over exchanges, some with inflated volume; test: rerun on a subset of venues
- **Small, clustered cross-section**: 25 assets in 3 markets are not exchangeable, and 3 clusters are too few for cluster-robust inference; report the asset bootstrap as conditional on independence, plus a within-market bootstrap and leave-one-market-out
- **Mistyped formulas**: check the constants of the estimator against the paper and run a unit test on simulated prices with a known spread

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Microstructure: Readings and Code

Advanced courses on quantinar.com

- Applied Computational Finance

Reading, code and data

- Textbooks: Hasbrouck (2007), Bouchaud, Bonart, Donier & Gould (2018), Harris (2003), Franke, Härdle & Hafner (2019)
- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AMM**: Automated Market Maker
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CARA**: Constant Absolute Risk Aversion
- **CEX**: Centralised EXchange
- **CFTC**: Commodity Futures Trading Commission
- **CI**: Confidence Interval
- **DEX**: Decentralised EXchange
- **EDGE**: Efficient Discrete Generalized Estimator (of the bid-ask spread)
- **EKOP**: Easley–Kiefer–O'Hara–Paperman (model of informed trading)
- **ETF**: Exchange-Traded Fund
- **FOK**: Fill-Or-Kill (order)
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **GMM**: Generalized Method of Moments
- **GTC**: Good-Till-Cancelled (order)

Acronyms used in this material (2/2)

- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HFT**: High-Frequency Trading
- **IOC**: Immediate-Or-Cancel (order)
- **LLM**: Large Language Model
- **LOB**: Limit Order Book
- **MRR**: Madhavan–Richardson–Roomans (structural model of trading costs)
- **NYSE**: New York Stock Exchange
- **OHLC**: Open, High, Low, Close
- **OLS**: Ordinary Least Squares
- **PIN**: Probability of INformed trading
- **SEC**: Securities and Exchange Commission
- **TWAP**: Time-Weighted Average Price
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error-Correction Model
- **VIX**: CBOE Volatility IndeX
- **VPIN**: Volume-synchronised Probability of INformed trading
- **VWAP**: Volume-Weighted Average Price
- **XRP**: XRP (the native token of the XRP Ledger)
- **TR**: Total Return

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